

| UCS551: Conversational AI: Accelerated Data Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   |    |
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| <b>Course Objective:</b> This course will provide students with fundamental and advanced methods of data science: from Data Collection to Analytics and Machine Learning with RAPIDS on Text and Graphical problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |   |    |
| <b>Syllabus</b> <p><b>Introduction:</b> Fundamentals of Data Science, GPU Acceleration, RAPIDS Framework</p> <p><b>Data Collection:</b> Collecting Data, Scraping Data, Popular Scraping libraries, Data Annotation</p> <p><b>Data Pre-processing (ETL):</b> Introduction to Data-preprocessing, Data Cleaning &amp; Statistical Preprocessing, Data Cleaners: OpenRefine and Wrangler, Feature Selection: Introduction to Filter Methods, Introduction to Model- based methods, Feature Reduction: PCA.</p> <p><b>Introduction to Machine Learning - Supervised:</b> Introduction to Supervised Learning, Linear Model, RAPIDS acceleration: Linear Regression, Overfitting and Cross Validation, Decision Tree, Visualizing Classification: {ROC, AUC, Confusion Matrix}, Bagging, Random Forests, RAPIDS Acceleration: Random Forest, Boosting, RAPIDS acceleration: K-NN, XGBoost.</p> <p><b>Introduction to Machine Learning - Unsupervised:</b> Introduction to Unsupervised Learning, Kmeans &amp; Hierarchical Clustering, RAPIDS acceleration: K-Means, DBSCAN, PCA, t-SNE, UMAP, Visualizing Clusters, RAPIDS acceleration: PCA [t-SNE], UMAP, DBSCAN.</p> <p><b>Graph Analytics:</b> How to Represent &amp; Store Graphs, Graph Power Laws, Centralities: Degree, Betweenness, Clustering Coefficient, PageRank &amp; Personalized PageRank, Interactive Graph Exploration, RAPIDS Acceleration: Graphistry &amp; cuXFilter.</p> <p><b>Introduction to Deep Learning:</b> Introduction to Artificial Neural Networks (ANNs), Artificial Neurons, Layers, Perceptron, Multilayer Perceptron, Advanced Deep Neural Networks (DNNs), Batch Normalization, Hyperparameter tuning, Activation Functions, Metrics, Optimization, Regularization.</p> |   |   |   |    |
| <b>Laboratory Work:</b> <ol style="list-style-type: none"> <li>1. Introduction to Dockers &amp; Containers, Introduction to NVIDIA GPU Cloud (NGC).</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |    |

2. Practical on Traditional Data Science packages (Numpy, Pandas, Scipy, Scikit-Learn).
3. Accelerated Data Science framework RAPIDS: Introduction to RAPIDS and cuDF.
4. Data Collection via API/Web Scraping.
5. Decision Tree Classification Clustering in RAPIDS.
6. Random Forest Classification in RAPIDS.
7. KMeans Clustering Implementation in RAPIDS.
8. Dimensionality Reduction and Visualization in RAPIDS.
9. Graph Analytics with cuGraph.
10. Latent semantic indexing for text via singular value decomposition(cuML).
11. Accelerating Workloads using RAPIDS
12. Introduction to DL Frameworks: PyTorch, and Tensorflow (Keras)

### **Course Learning Objectives (CLO)**

After the completion of the course the student will be able to:

1. In depth understanding of Data Analytics, Pre-processing and Visualization Toolkits.
2. Comprehend and apply different classification and clustering techniques.
3. Understand GPU computing for building advanced data science applications.
4. Understand the concept of Neural Networks and its implementation using deep learning frameworks.

### **Text Books**

1. Mitchell M., T., *Machine Learning*, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., *Introduction to Machine Learning*, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, *Machine Learning with Python*, BPB Publication (2018).

### **Reference Books:**

1. Bishop M., C., *Pattern Recognition and Machine Learning*, Springer-Verlag (2011) 2nd Edition.
2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., *Machine Learning, Neural and Statistical Classification*. Overseas Press (1994).